Software Requirements Specification

Community Residual Waste Management System

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	Details
Team name	Larb Lamphon
Project title	Community Residual Waste Management System — Citizen Reporting and Collection Tracking
SDLC chosen in M1 + reason	Incremental- Once feedback is received, it can be used for ongoing tracking and improvement.
Date / version	9 Oct 2026 · v1.0
GitHub repo	https://github.com/Adichat-dev/Larb-Lamphoon.git

1. Introduction

1.1 Purpose and scope

Residents who find overflowing bins or illegally dumped waste in their community have no clear way to tell the municipality, and no way to know whether anyone has acted on the problem. Collection staff, in turn, receive complaints in scattered channels with no consistent priority or status. The Community Residual Waste Management System is a web application that lets citizens report a waste spot with a type, urgency, location and photo, receive a tracking number, and follow the report until the waste is collected. Collection officers see a queue sorted by urgency and update each job's status, and municipality administrators see summary figures, areas with repeated problems, and a CSV report. The MVP covers reporting, tracking, status updates, status-change notification, and summary reporting. Success looks like a citizen can see the current status of any report from its tracking number, and an administrator can see which areas have repeated problems without compiling them by hand.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Citizen (primary user)	primary user	Report a waste spot with photo and location, get a tracking number, and see progress	Reports being ignored or lost, with no way to know the status
Collection officer (primary user)	primary user	See pending jobs ordered by urgency, accept a job, and mark it collected with an optional after-photo	Unclear priorities, missing location details, and duplicate or stale jobs
Municipality administrator	secondary stakeholder	View totals by status, find areas with repeated reports, and export a report	Recurring problem areas going unnoticed; having no evidence for planning or reporting

2. Overall description

2.1 Product context — what it does / does not

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 Report waste spot with photo / location • FR-2 Receive tracking number and track status • FR-3 Officer views queue and updates status • FR-4 Notify on status change (simulated) • FR-5 Summary dashboard and CSV report	• Real SMS / LINE / e-mail push notifications • User accounts and login • Truck routing or collection scheduling • Waste treatment, recycling or disposal processing

2.2 Assumptions and constraints

Assumption: The prototype uses simulated data only; real reports will be entered by citizens using a smartphone or computer browser with an internet connection.

Constraint: The prototype stores all data in the user's own browser (local storage), so data is not shared between devices.

Constraint: Notifications are simulated as on-screen messages and are not sent to citizens' phones.

Constraint: The user interface is in Thai; this SRS is written in English.

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Every story traces to a pain, and every acceptance criterion is testable.

FR-1 — Report waste spot with photo and location

User story: As a citizen, I want to report a waste spot by choosing a waste type and urgency, describing the problem, and giving a location and an optional photo, so that the municipality knows exactly where and what the problem is.

Pain this traces to: residents have no single, structured way to report waste, and reports lack a clear location and evidence.

Acceptance criteria:

1. Given the description and location are filled in, when the citizen submits the report, then the system creates a report with the chosen type and urgency and the status "Pending".

2. Given the description or location is empty, when the citizen submits, then the system rejects the report and asks for the missing information.

3. Given the citizen attaches an image, when the report is submitted, then the photo is saved with the report.

FR-2 — Receive tracking number and track status

User story: As a citizen, I want to receive a tracking number and look up my report with it, so that I can see its current status and history without contacting anyone.

Pain this traces to: citizens do not know whether a report was received or acted on.

Acceptance criteria:

1. Given a report is submitted, when it is saved, then the system shows a unique tracking number in the format WS-0001.

2. Given a valid tracking number, when the citizen searches, then the system shows the current status, urgency, type, location, description, photos and a dated status history.

3. Given an unknown tracking number, when the citizen searches, then the system shows a "not found" message.

FR-3 — Officer views queue and updates status

User story: As a collection officer, I want to see open jobs ordered by urgency and update each job's status, so that I handle the most urgent waste first and everyone can see progress.

Pain this traces to: officers receive complaints without priority or status, so urgent problems wait behind routine ones.

Acceptance criteria:

1. Given open reports exist, when the officer opens the job list, then reports with status Pending or In progress are listed with the most urgent first and, within the same urgency, the oldest first.

2. Given a Pending job, when the officer accepts it, then its status becomes "In progress" and the change is added to the history with a time stamp.

3. Given an In progress job, when the officer marks it collected (optionally attaching an after-photo), then its status becomes "Collected" and the job leaves the open list.

FR-4 — Notify on status change

User story: As a citizen, I want to be notified when my report's status changes, so that I know when action has been taken.

Pain this traces to: citizens must keep checking back to find out whether anything has happened.

Acceptance criteria:

1. Given an officer changes a report's status, when the change is saved, then the system displays a notification naming the tracking number and the new status.

2. Given a status change, when the citizen later looks up the tracking number, then the new status and time appear in the history.

FR-5 — Summary dashboard and CSV report

User story: As a municipality administrator, I want to see totals by status, areas with repeated reports, and export the reports as a CSV file, so that I can plan work and report results.

Pain this traces to: recurring problem areas are not noticed because reports are not summarised anywhere.

Acceptance criteria:

1. Given reports exist, when the administrator opens the dashboard, then it shows the total number of reports and the number in each status.

2. Given the same location was reported more than once, when the administrator opens the dashboard, then that location is listed with its number of reports.

3. Given reports exist, when the administrator exports, then a CSV file is downloaded with tracking number, area, waste type, urgency, description, status and report time for every report.

4. Non-functional requirements

Statements are measurable; at least one covers security / privacy.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	Every user action (submit report, search, change status, switch role view, export CSV) shall complete and update the screen within 10 seconds.	Built-in response-time panel, 3 trials per action
NFR-2	usability	A first-time citizen shall submit a complete report in under 90 seconds without help from a teammate.	Timed user test with 3 people

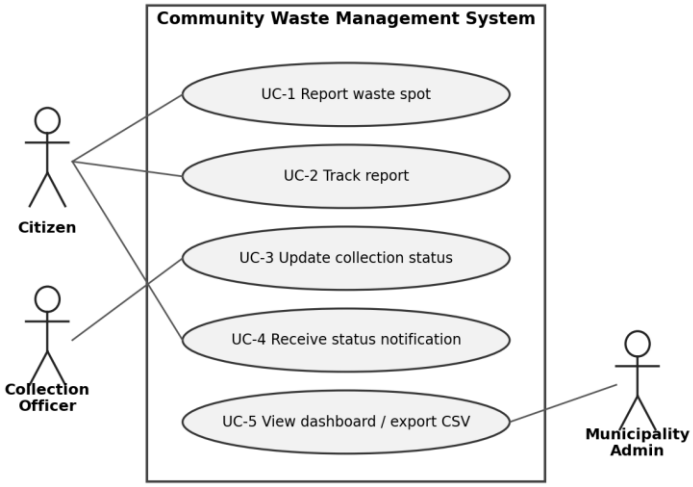
ID	Kind	Statement (testable)	How we will check
NFR-3	reliability	Submitted reports and their status history shall still be present after the user refreshes or closes and reopens the page.	Refresh and reopen test
NFR-4	security / privacy	The citizen view shall show a report only when its exact tracking number is entered, and shall not list other citizens' reports or personal data.	Search with a wrong number; inspect the citizen view
NFR-5	compatibility	All screens shall be usable without horizontal scrolling on a 360-pixel-wide phone screen and on a desktop browser.	Resize browser to 360 px; test on one phone

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1	Citizen	FR-1	Citizen fills in type, urgency, description, location and photo and submits a waste report.
UC-2	Citizen	FR-2	Citizen enters a tracking number and sees the report's status and history.
UC-3	Collection officer	FR-3	Officer accepts a job and later marks it collected with an after-photo.
UC-4	Citizen / Collection officer	FR-4	A status change is saved and a notification showing the new status appears.
UC-5	Municipality administrator	FR-5	Administrator reviews totals and repeated areas and exports a CSV report.

5.2 Use-case diagram



6. Out of scope

1. Real SMS, LINE or e-mail push notifications (notifications are simulated on screen).
2. User accounts, login and role-based access control.
3. Truck routing, shift planning or collection scheduling.
4. Waste treatment, recycling programmes or disposal processing.
5. Fines or enforcement against people who dump waste.

7. Traceability matrix (Golden Thread)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Reports lack a structured form, location and evidence	FR-1 report with photo / location	Report form with type, urgency, location, photo	Report form and GPS button	Submit-report test
Citizens cannot tell if a report was received or acted on	FR-2 tracking number	Ticket ID and history timeline	Tracking search (WS-0001)	Tracking lookup test
Urgent waste waits behind routine complaints	FR-3 officer queue and status	Urgency-sorted job list with status buttons	Officer job list, after-photo	Queue order and status test
Citizens must keep checking for progress	FR-4 status notification	Notification on status change	On-screen notification	Notification test
Repeated problem areas go unnoticed	FR-5 dashboard and CSV	Counts by status, repeated-area list, CSV export	Admin dashboard and export	Dashboard and CSV test

8. AI usage log

Log every prompt that drafted a story or paragraph, honestly.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
22 Oct 2026	Claude	Write an English SRS in the format of the Word template, using the uploaded HTML prototype for the content.	Drafted all sections: scope, stakeholders, FRs, NFRs, use cases, diagram, traceability and out-of-scope content.	Team must check team name, SDLC, repo, NFR numbers and the pain statements against real users and the prototype.	Each FR should be traced to the prototype's features and to a pain that is verified with real residents.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned

9. Self-check

☐ Every FR is a user story with a benefit

☐ Every FR has testable acceptance criteria

☐ No FR is vague (no bare "fast", "easy", "user-friendly")

☐ NFRs exist, including security / privacy

☐ Every FR traces back to a stated problem

☐ Scope and out-of-scope are stated

☐ Use-case list matches the FR list

☐ AI log is honest